

Week – 4 Assignment

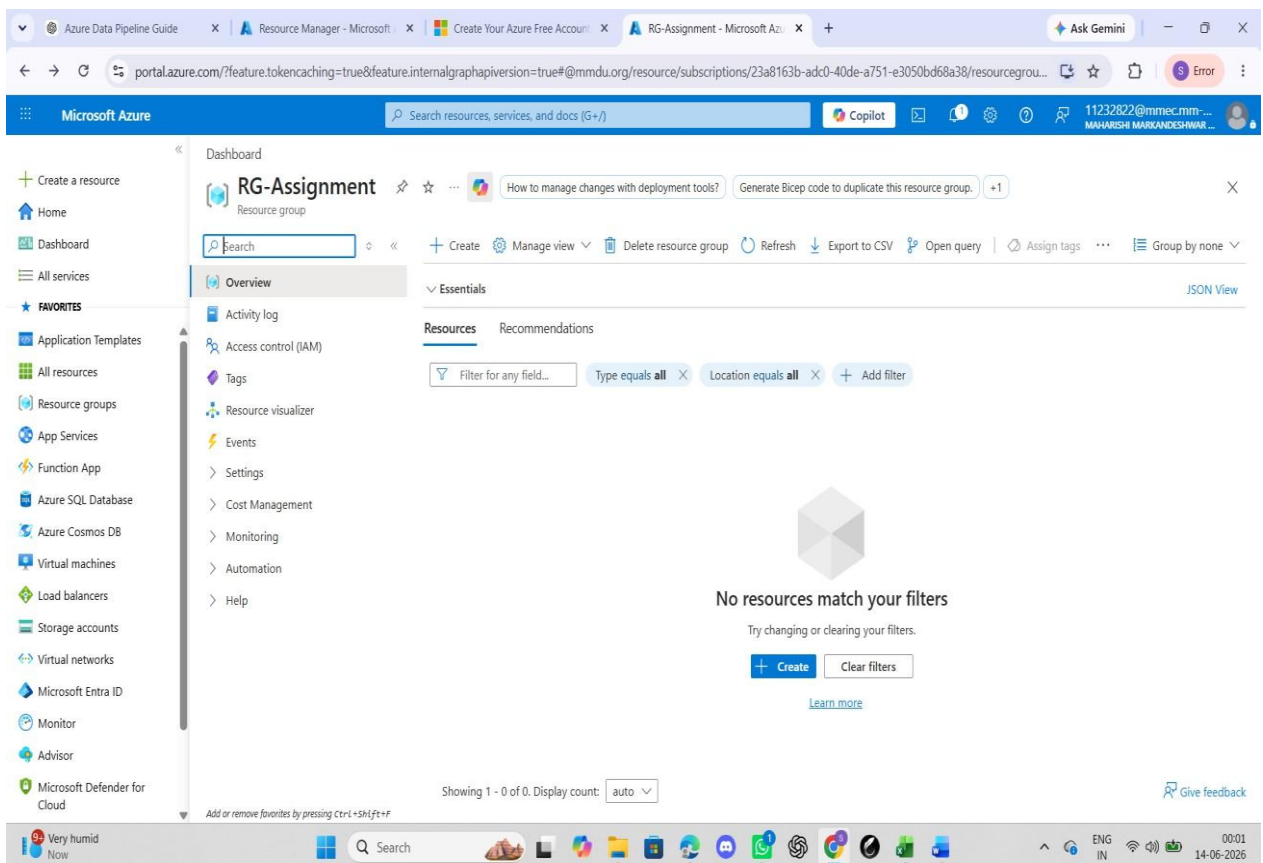
Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Submitted By: Shivam Kumar Rajput (MMDU)

1. Objective

Understand Azure cloud concepts and build an end-to-end data pipeline using Storage Account and Azure Data Factory. Steps: Explore Azure Portal and create a Resource Group. Create a Storage Account, Blob Container, and upload a CSV file. Create Azure Data Factory and explore ADF UI (Author, Monitor, Manage). Create Linked Service (Blob Storage) and datasets (source and destination). Use Get Metadata activity to retrieve file information. Build a pipeline using Copy Data activity and configure source and destination. Execute the pipeline using Debug/Trigger and monitor execution. Assign IAM roles (Reader, Contributor) and ensure access between ADF and Storage. Implement an end-to-end pipeline (Blob → ADF → Destination) with metadata validation.

2. Resource Group Creation



3. Storage Account Creation

The screenshot shows the 'Create a storage account' wizard in the Microsoft Azure portal. The 'Review + create' tab is active, showing the following configuration details:

Section	Parameter	Value
Basics	Subscription	Azure subscription 1
	Resource group	RG-Assignment
	Location	Central India
	Storage account name	sameepstorage822
	Primary service	Standard
Performance	Performance	Standard
	Replication	Read-access geo-redundant storage (RA-GRS)
Advanced	Enable hierarchical namespace	Disabled
	Enable SFTP	Disabled
	Enable network file system v3	Disabled
	Allow cross-tenant replication	Disabled
	Access tier	Hot
	Managed Identity for CMR	Disabled

The 'Create' button is visible at the bottom of the wizard. The left sidebar shows the 'Storage accounts' option under 'All services'.

4. Blob Container Creation

The screenshot shows the 'New container' wizard in the Microsoft Azure portal. The 'Name' field is set to 'input', and the 'Anonymous access level' is set to 'Private (no anonymous access)'. The 'Create' button is visible at the bottom of the wizard. The left sidebar shows the 'Containers' option under 'Data storage'.

The 'New container' wizard displays the following configuration details:

Field	Value
Name *	input
Anonymous access level	Private (no anonymous access)

The 'Create' button is visible at the bottom of the wizard. The left sidebar shows the 'Containers' option under 'Data storage'.

5. CSV File Upload

Dashboard > Storage center | Blob Storage > sameepstorage822 | Containers

input
Container

Search resources, services, and docs (G+)

Overview

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - Superstore.csv	6/14/2026, 12:08:27 AM	Hot (Inferred)	Block blob	2.18 MiB	Available

6. Azure Data Factory Creation

Dashboard > Data factories

Create Data Factory

Basics | Git configuration | Networking | Advanced | Tags | Review + create

View automation template

Basics

Subscription: Azure subscription 1
Resource group: Sameep-assignment
Region: Central India
Version: V2

Networking

Connect via: Public endpoint

Previous | Next | Create

Dashboard > CreateDataFactory-20260614001047 | Overview

Deployment

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name: CreateDataFactory-20260614001047... Start time: 6/14/2026, 12:12:33 AM
Subscription: Azure subscription 1 Correlation ID: 0775685b-39aa-40d4-859b-b3a66...

Deployment details

Next steps

Go to resource

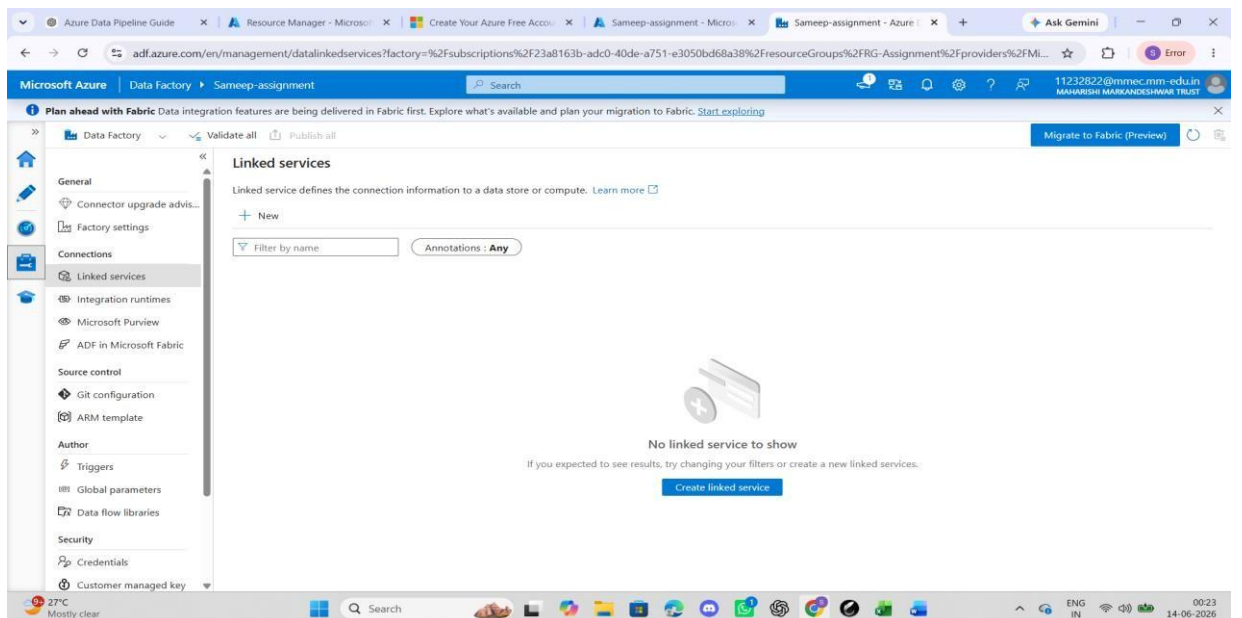
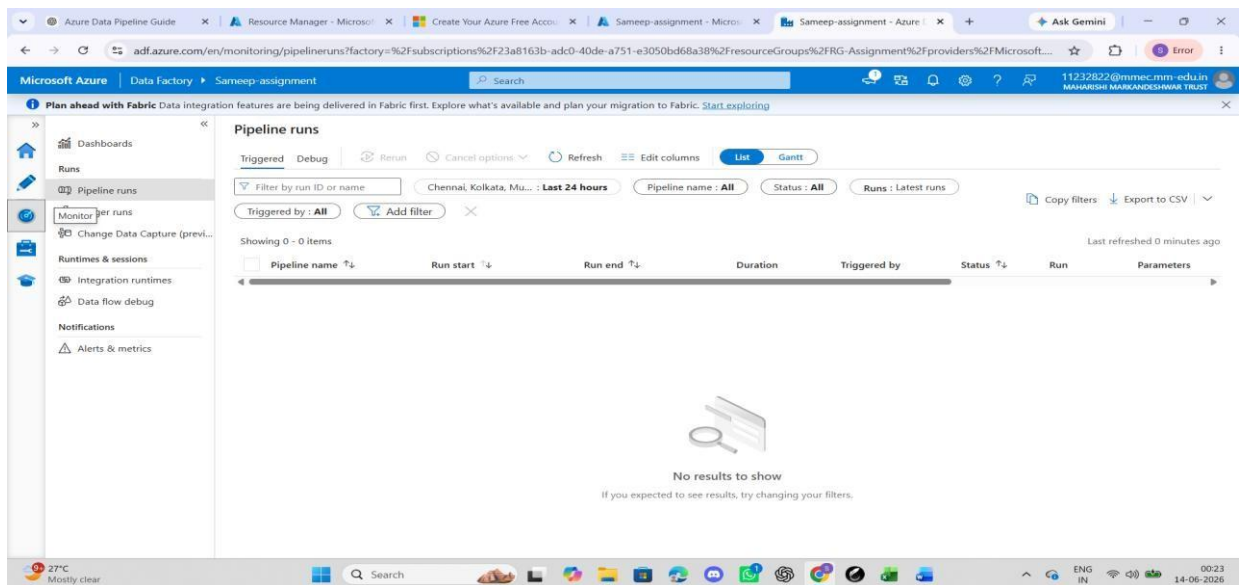
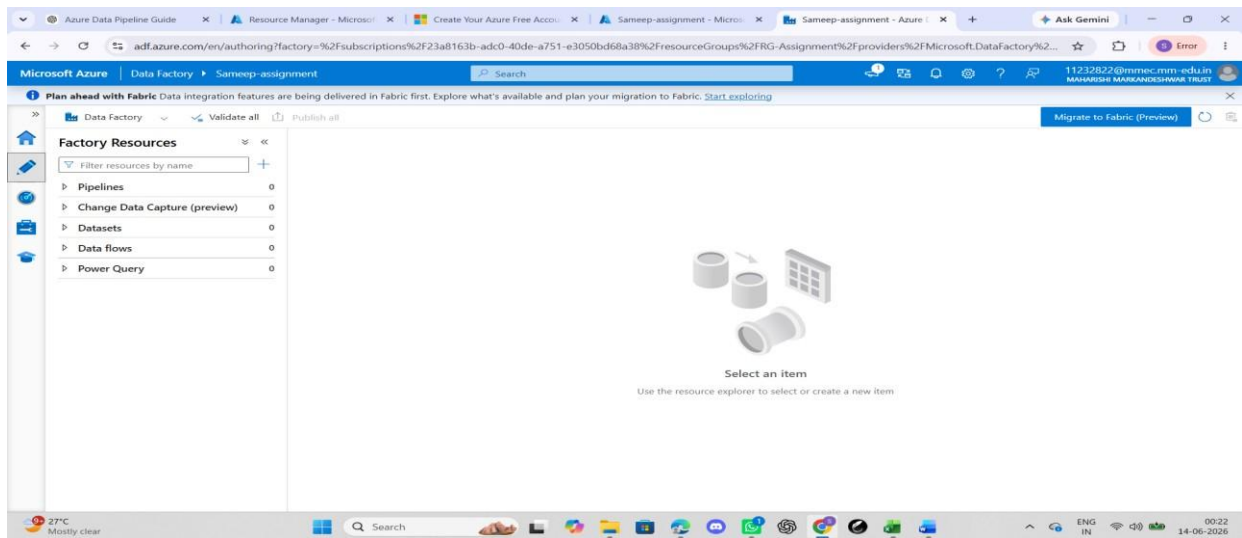
Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >

Microsoft Defender for Cloud
Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >

Free Microsoft tutorials
Start learning today >

Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
Find an Azure expert >

7. ADF Studio Overview



8. Linked Service Configuration

The screenshot shows the Microsoft Azure Data Factory portal interface. The left sidebar contains navigation options like General, Connections, Integration runtimes, and Source control. The main area displays the 'Linked services' section with a 'New' button. A modal window titled 'New linked service' is open, showing configuration for 'LS_BlobStorage'. The configuration includes fields for Name, Description, Connect via integration runtime (set to 'AutoResolveIntegrationRuntime'), Authentication type (set to 'Account key'), Account selection method (set to 'From Azure subscription'), Azure subscription (set to 'Select all'), and Storage account name (set to 'sameepstorage822'). There are also buttons for 'Create', 'Back', 'Test connection', and 'Cancel'.

9. Source Dataset Creation

The screenshot shows the Microsoft Azure Data Factory portal interface. The left sidebar contains navigation options like Pipelines, Datasets, Data flows, and Power Query. The main area displays the 'Factory Resources' section with a 'DelimitedText1' dataset. A modal window titled 'DelimitedText1' is open, showing configuration for the dataset. The configuration includes fields for Name (set to 'DelimitedText1'), Description, and Annotations. The 'Connection' tab is selected, showing fields for Linked service (set to 'LS_BlobStorage'), File path (set to 'input / Directory / Sample - Superstore.csv'), Compression type (set to 'No compression'), Column delimiter (set to 'Comma (,)'), Row delimiter (set to 'Default (\r\n, or \n)'), Encoding (set to 'Default(UTF-8)'), and Quote character (set to 'Double quote (")'). There are also buttons for 'Test connection', 'Edit', 'New', 'Learn more', and 'Cancel'.

10. Pipeline Creation

The screenshot displays the Microsoft Azure Data Factory portal interface. The top navigation bar shows the user is logged in as '11232822@mimcc-mm-edu.in'. The main content area is titled 'Sameep-assignment' and shows a pipeline named 'PL_CopyData'. The left sidebar lists 'Factory Resources' including Pipelines, Datasets, Data flows, and Power Query. The 'Activities' pane on the right shows a list of activities: Move and transform, Synapse, Azure Data Explorer, Azure Function, Batch Service, Databricks, General, HDInsight, Iteration & conditionals, Machine Learning, and Power Query. The 'Properties' pane on the far right shows the 'General' tab for the 'PL_CopyData' pipeline, with fields for Name and Description. The 'Parameters' tab is also visible, showing a 'New' button. The bottom status bar indicates the time is 00:33 on 14-06-2026.

11. Get Metadata Activity

The screenshot displays the Microsoft Azure Data Factory portal interface, focusing on the 'Get Metadata' activity configuration. The top navigation bar shows the user is logged in as '11232822@mimcc-mm-edu.in'. The main content area is titled 'Sameep-assignment' and shows a pipeline named 'PL_CopyData'. The left sidebar lists 'Factory Resources' including Pipelines, Datasets, Data flows, and Power Query. The 'Activities' pane on the right shows a list of activities: Get, General, and Get Metadata. The 'Properties' pane on the far right shows the 'General' tab for the 'PL_CopyData' pipeline, with fields for Name and Description. The 'Settings' tab is selected, showing the 'Dataset' dropdown set to 'DelimitedText1' and the 'Field list' section with checkboxes for 'Argument', 'Exists', and 'Size'. The bottom status bar indicates the time is 00:36 on 14-06-2026.

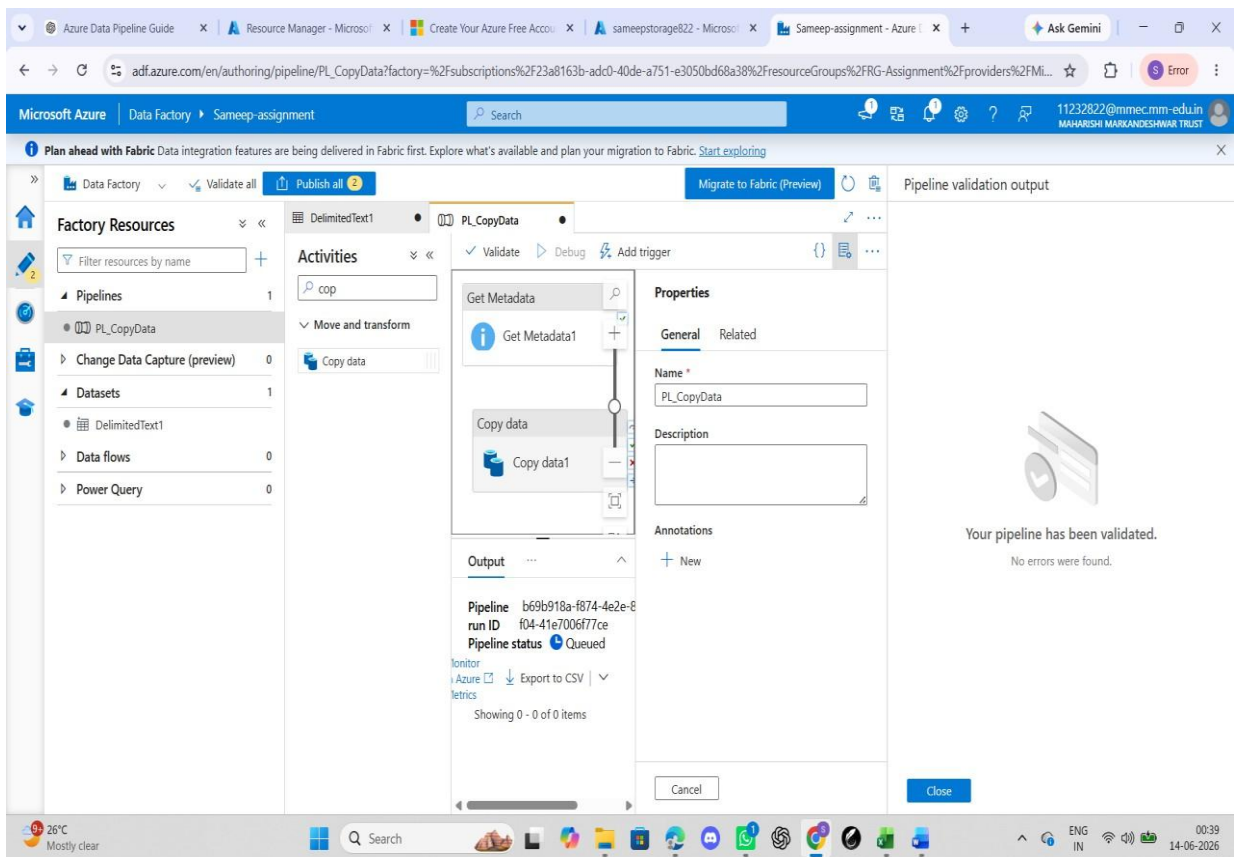
12. Copy Data Activity

The screenshot displays the Microsoft Azure Data Factory portal interface. The top navigation bar includes the Microsoft Azure logo, the current Data Factory name 'Sameep-assignment', and a search bar. A notification banner at the top mentions 'Plan ahead with Fabric Data integration features'. The left sidebar shows the 'Factory Resources' tree with categories like Pipelines, Datasets, Data flows, and Power Query. The main workspace is titled 'PL_CopyData' and shows a pipeline diagram with two activities: 'Get Metadata' and 'Copy data'. The 'Copy data' activity is selected, and its 'Settings' tab is active, showing 'Concurrency' and 'Elapsed time metric' options. The right sidebar displays the 'Properties' panel for the selected activity, with fields for 'Name' (PL_CopyData) and 'Description'. The bottom status bar shows the system clock and weather information.

13. Validate

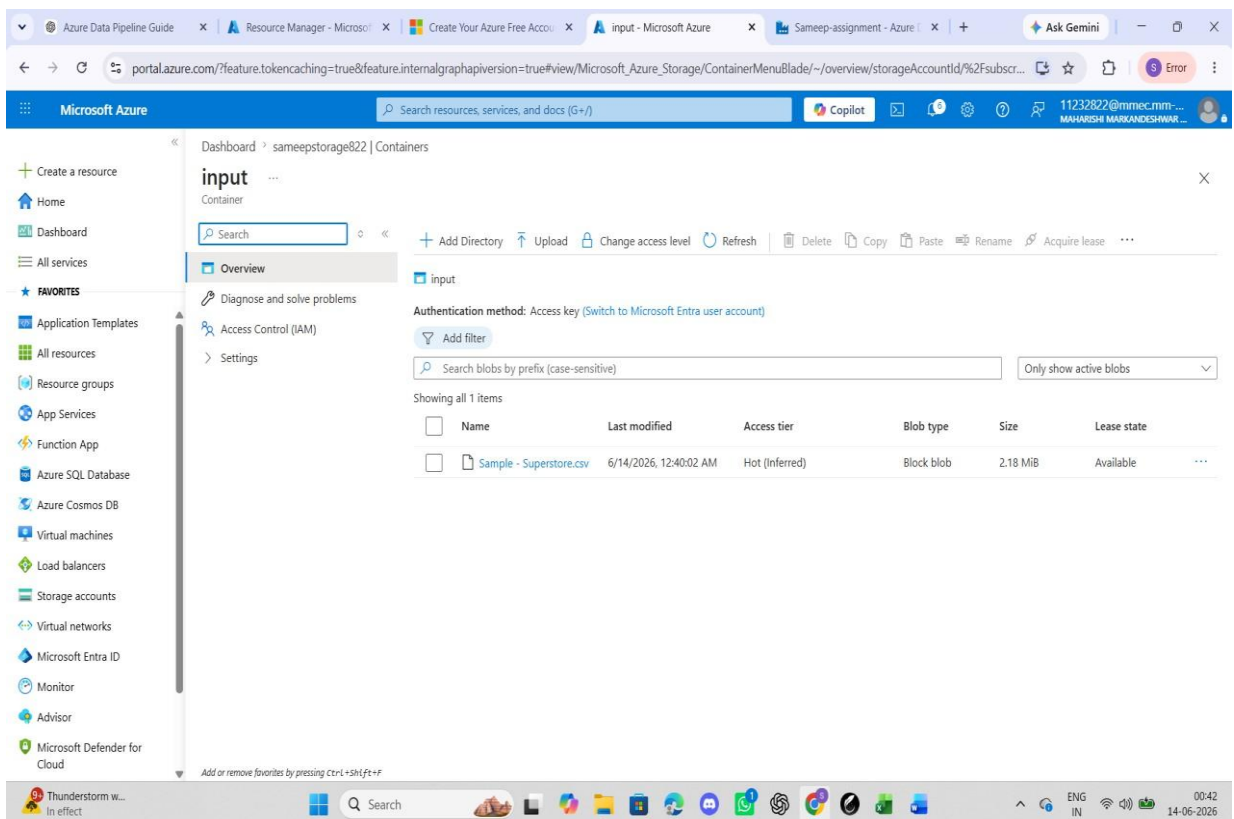
This screenshot shows the same Microsoft Azure Data Factory portal as the previous image, but with the 'Validate' button in the top navigation bar highlighted. The pipeline diagram in the main workspace remains the same. The right sidebar now displays the 'Pipeline validation output' panel, which contains a large green checkmark icon and the text: 'Your pipeline has been validated. No errors were found.' The bottom status bar shows the system clock and weather information.

14. Debug



The screenshot shows the Azure Data Factory portal interface. The left sidebar displays the 'Factory Resources' tree with 'Pipelines' expanded, showing 'PL_CopyData'. The main area shows the pipeline canvas with 'Get Metadata1' and 'Copy data1' activities. The 'Properties' pane on the right shows the pipeline's name 'PL_CopyData' and its status 'Queued'. The 'Output' pane below the canvas shows pipeline details: Pipeline ID 'b69b918a-f874-4e2e-8f04-41e7006f77ce', Pipeline status 'Queued', and a 'Monitor' button. The rightmost pane shows the 'Pipeline validation output' with a message: 'Your pipeline has been validated. No errors were found.'

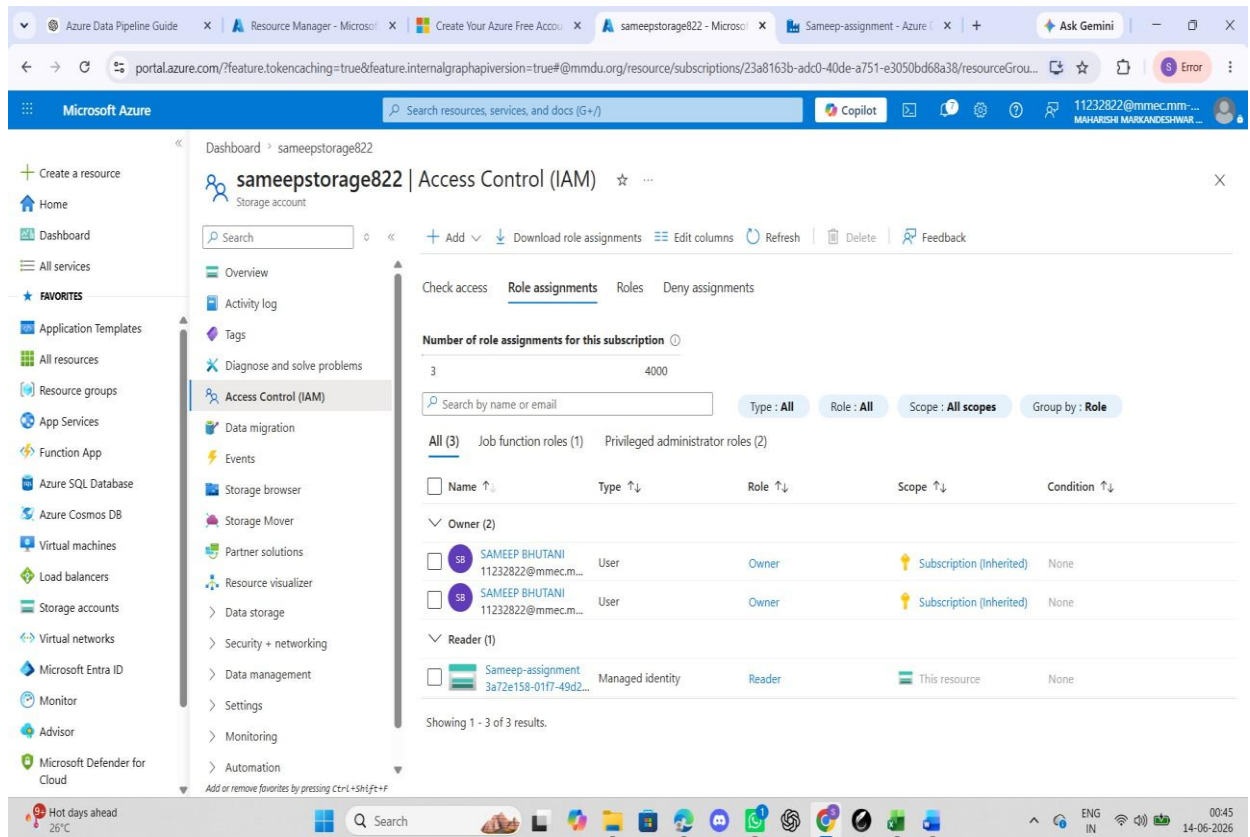
15. Output Check



The screenshot shows the Azure portal interface. The left sidebar displays the 'Microsoft Azure' navigation pane with 'Containers' expanded. The main area shows the 'input' container under 'sameepstorage822'. The 'Overview' tab is selected, showing a table with one item: 'Sample - Superstore.csv'. The table columns are Name, Last modified, Access tier, Blob type, Size, and Lease state.

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - Superstore.csv	6/14/2026, 12:40:02 AM	Hot (Inferred)	Block blob	2.18 MiB	Available

16. IAM Role Assignment



17. Result

The pipeline executed successfully and copied the CSV file from the source container to the destination container.

18. Conclusion

This assignment successfully demonstrated the implementation of an end-to-end data pipeline using Microsoft Azure services. A Resource Group, Storage Account, Blob Containers, and Azure Data Factory were created and configured. The provided CSV file was uploaded to Azure Blob Storage and connected to Azure Data Factory through Linked Services and Datasets. The Get Metadata activity was used to validate and retrieve information about the source file, while the Copy Data activity was used to transfer the file from the source container to the destination container. The pipeline was executed successfully using the Debug and Trigger options, and the execution status was monitored through the Monitor section of Azure Data Factory. Proper access permissions were also ensured through IAM role assignments. The successful creation and execution of the pipeline confirmed the seamless integration between Azure Storage and Azure Data Factory for data movement and processing.